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example, median survival among individuals with CF is <30 years in much of Latin America (compared to >45 years in the United States). The less favorable prognosis is attributable in part to lack of widespread diagnostic capabilities (i.e., newborn screening, sweat testing, and genetic analysis tailored to ethnic background), together with insufficient access to leading-edge, interdisciplinary treatment. Efforts to apply state-of-the-art management to underdiagnosed and underserved CF patient populations will help improve outcomes and mitigate CF health disparities in the future.

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VIDEO 291-1 Role of CFTR during airway mucociliary clearance. Initial video sequences depict establishment of the normal periciliary fluid layer bathing the surface airway epithelium, with spheres representing chloride and bicarbonate ions secreted through CFTR and across the apical (mucosal) respiratory surface. Later video describes failure of CFTR anion transport and resulting depletion of the periciliary layer, "plastering" of cilia against the mucosal surface, and accumulation of mucus in the airway with resulting bacterial infection. *(Reproduced with permission from Cystic Fibrosis Foundation.)*

VIDEO 291-2AB Pharmacologic modulation of mutant *CFTR*. Initial video (***A***) illustrates CFTR encoding an ion transport gating (class III) defect. The CF gene product is localized to the plasma membrane but incapable of conducting anions (*yellow spheres*) until a potentiator molecule (shown in *green*) binds and facilitates channel opening. Later video (***B***) describes CFTR encoding a maturational processing (protein biogenesis, class II) defect. The mutant protein is misfolded, fails to traffic to the cell surface, and is degraded by the proteasome. Binding of corrector molecules (*red spheres*) improves folding and facilitates CFTR stabilization and cell surface localization/function. *(Reproduced with permission from Cystic Fibrosis Foundation.)*

292 Chronic Obstructive Pulmonary Disease

Edwin K. Silverman, James D. Crapo, Barry J. Make

Chronic obstructive pulmonary disease (COPD) is defined as a disease state characterized by persistent respiratory symptoms and airflow obstruction (*https://goldcopd.org/2021-gold-reports/*). COPD includes *emphysema*, an anatomically defined condition characterized by destruction of the lung alveoli with air space enlargement; *chronic bronchitis*, a clinically defined condition with chronic cough and phlegm; and/or *small airway disease*, a condition in which small bronchioles are narrowed and reduced in number. The *classic definition* of COPD requires the presence of chronic airflow obstruction, determined by spirometry, that usually occurs in the setting of noxious environmental exposures—most commonly products of combustion, cigarette smoking in the United States, and biomass fuels in some other countries. Host factors such as abnormal lung development and genetics can lead to COPD. Emphysema, chronic bronchitis, and small airway disease are present in varying degrees in different COPD patients. Patients with a history of cigarette smoking without chronic airflow obstruction may have chronic bronchitis, emphysema, and dyspnea. Although these patients are not included within the classic definition of COPD, they may have similar disease processes. Respiratory symptoms and other features of COPD can occur in subjects who do not meet a definition of COPD based only on airflow obstruction determined by spirometric population thresholds of normality. Investigators in the COPDGene study recently proposed a multidimensional approach to COPD diagnosis, which is based on domains of environmental exposures, respiratory symptoms, imaging abnormalities, and physiologic abnormalities.

COPD is the fourth leading cause of death and affects >10 million persons in the United States. COPD is also a disease of increasing public health importance around the world. Globally, there are an estimated 250 million individuals with COPD.

PATHOGENESIS

Airflow obstruction, the physiologic marker of COPD, can result from airway disease and/or emphysema. Small airways may become narrowed by cells (hyperplasia and accumulation), mucus, and fibrosis, and extensive small airway destruction has been demonstrated to be a hallmark of COPD. Although the precise biological mechanisms leading to COPD have not been determined, a number of key cell types, molecules, and pathways have been identified from cell-based and animal model studies. The pathogenesis of emphysema (shown in Fig. 292-1) is more clearly defined than the pathogenesis of small airway disease. Pulmonary vascular destruction occurs in concert with small airway disease and emphysema.

The current dominant paradigm for the pathogenesis of emphysema comprises a series of four interrelated events: (1) Chronic exposure to cigarette smoke in genetically susceptible individuals triggers inflammatory and immune cell recruitment within large and small airways and in the terminal air spaces of the lung. (2) Inflammatory cells release proteinases that damage the extracellular matrix supporting airways, vasculature, and gas exchange surfaces of the lung. (3) Structural cell death occurs through oxidant-induced damage, cellular senescence, and proteolytic loss of cellular-matrix attachments leading to extensive loss of smaller airways, vascular pruning, and alveolar destruction. (4) Disordered repair of elastin and other extracellular matrix components contributes to air space enlargement and emphysema.

INFLAMMATION AND EXTRACELLULAR MATRIX PROTEOLYSIS

Elastin, the principal component of elastic fibers, is a highly stable component of the extracellular matrix that is critical to the integrity of the lung. The elastase:antielastase hypothesis, proposed in the mid-1960s, postulated that the balance of elastin-degrading enzymes and their inhibitors determines the susceptibility of the lung to destruction, resulting in air space enlargement. This hypothesis was based on the clinical observation that patients with genetic deficiency in α_1 antitrypsin (α_1AT), the inhibitor of the serine proteinase neutrophil elastase, were at increased risk of emphysema, and that instillation of elastases, including neutrophil elastase, into experimental animals results in emphysema. The elastase:antielastase hypothesis remains a prevailing mechanism for the development of emphysema. However, a complex network of immune and inflammatory cells and additional biological mechanisms that contribute to emphysema have subsequently been identified. Upon exposure to oxidants from cigarette smoke, lung

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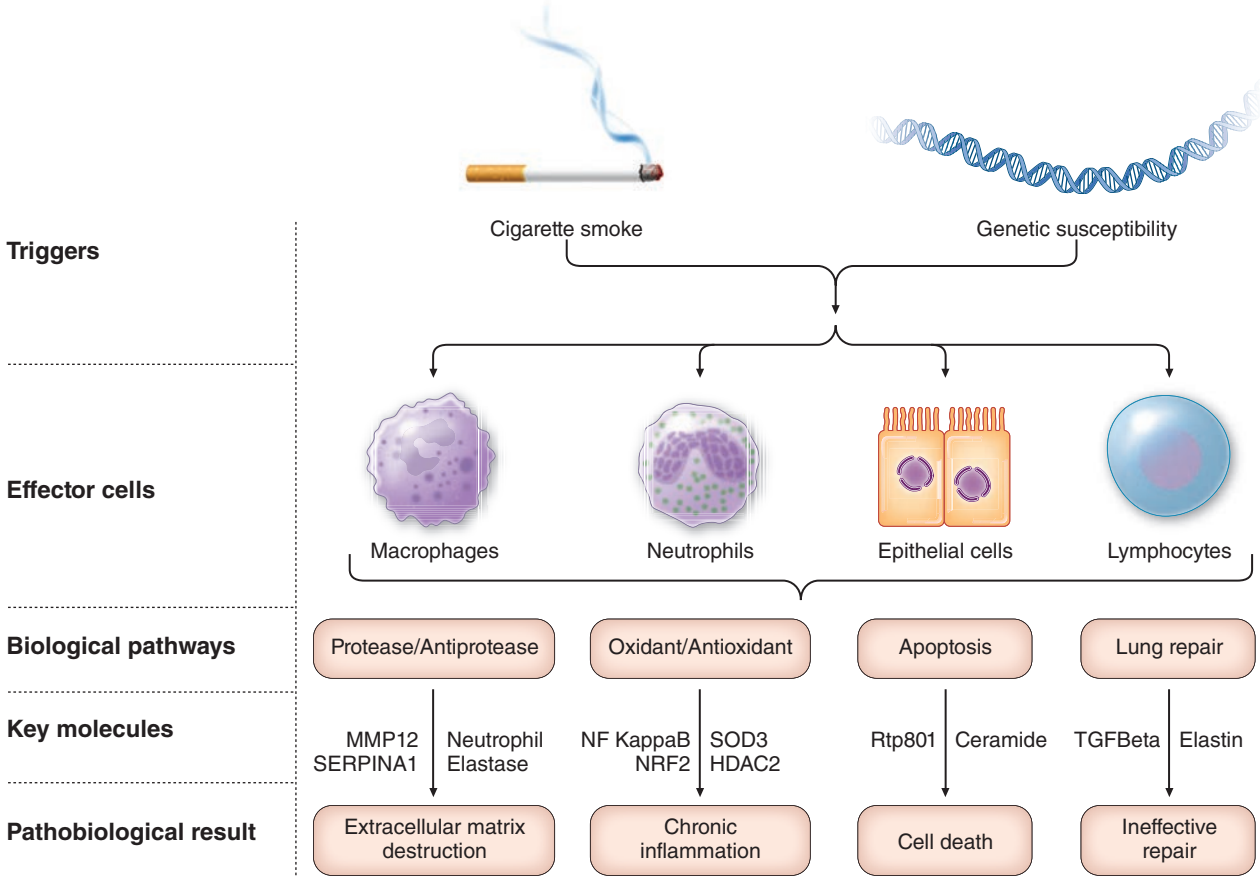


FIGURE 292-1 Pathogenesis of emphysema. Upon long-term exposure to cigarette smoke in genetically susceptible individuals, lung epithelial cells and T and B lymphocytes recruit inflammatory cells to the lung. Biological pathways of protease-antiprotease imbalance, oxidant/antioxidant imbalance, apoptosis, and lung repair lead to extracellular matrix destruction, cell death, chronic inflammation, and ineffective repair. Although most of these biological pathways influence multiple pathobiological results, only a single relationship between pathways and results is shown. A subset of key molecules related to these biological pathways is listed.

macrophages and epithelial cells become activated, producing proteinases and chemokines that attract other inflammatory and immune cells. Oxidative stress is a key component of COPD pathobiology; the transcription factor NRF2, a major regulator of oxidant-antioxidant balance, and SOD3, a potent antioxidant, have been implicated in emphysema pathogenesis by animal models. Mitochondrial dysfunction in COPD may worsen oxidative stress. One mechanism of macrophage activation occurs via oxidant-induced inactivation of histone deacetylase-2 (HDAC2), shifting the balance toward acetylated or open chromatin, exposing nuclear factor-κB sites, and resulting in transcription of matrix metalloproteinases and proinflammatory cytokines such as interleukin 8 (IL-8) and tumor necrosis factor α (TNF-α); this leads to neutrophil recruitment. CD8+ T cells are also recruited in response to cigarette smoke and release interferon-inducible protein-10 (IP-10, CXCL-7), which in turn leads to macrophage production of macrophage elastase (matrix metalloproteinase-12 [MMP-12]).

Matrix metalloproteinases and serine proteinases, most notably neutrophil elastase, work together by degrading the inhibitor of the other, leading to lung destruction. Proteolytic cleavage products of elastin serve as a macrophage chemokine, and proline-glycine-proline (generated by proteolytic cleavage of collagen) is a neutrophil chemokine—fueling this destructive positive feedback loop. Elastin degradation and disordered repair are thought to be primary mechanisms in the development of emphysema.

There is some evidence that autoimmune mechanisms may promote the progression of disease. Increased B cells and lymphoid follicles are present around the airways of COPD patients, particularly those with advanced disease. Antibodies have been found against elastin fragments as well; IgG autoantibodies with avidity for pulmonary epithelium and the potential to mediate cytotoxicity have been detected.

Concomitant cigarette smoke–induced loss of cilia in the airway epithelium and impaired macrophage phagocytosis predispose to bacterial infection with neutrophilia. In end-stage lung disease, long after smoking cessation, there remains an exuberant inflammatory response, suggesting that cigarette smoke–induced inflammation both initiates the disease and, in susceptible individuals, establishes a chronic process that can continue disease progression even after smoking cessation.

Cell Death Cigarette smoke oxidant-mediated structural cell death occurs via a variety of mechanisms including excessive ceramide production and Rtp801 inhibition of mammalian target of rapamycin (mTOR), leading to cell death as well as inflammation and proteolysis. Involvement of mTOR and other cellular senescence markers has led to the concept that emphysema resembles premature aging of the lung. Heterozygous gene-targeting of one of the leading genetic determinants of COPD identified by genome-wide association studies (GWAS), hedgehog interacting protein (*HHIP*), in a murine model leads to aging-related emphysema.

Ineffective Repair The ability of the adult lung to replace lost smaller airways and microvasculature and to repair damaged alveoli appears limited. Uptake of apoptotic cells by macrophages normally results in production of growth factors and dampens inflammation, promoting lung repair. Cigarette smoke impairs macrophage uptake of apoptotic cells, limiting repair. It is unlikely that the intricate and dynamic process of septation that is responsible for alveologenesis during lung development can be reinitiated in the adult human lung.

PATHOLOGY

Cigarette smoke exposure may affect the large airways, small airways (≤2 mm diameter), and alveoli. Changes in large airways cause cough and sputum production, while changes in small airways and alveoli are responsible for physiologic alterations. Airway inflammation,

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TABLE 292-1 GOLD Criteria for Severity of Airflow Obstruction in COPD

GOLD STAGE	SEVERITY	SPIROMETRY
I	Mild	FEV_1/FVC <0.7 and FEV_1 ≥80% predicted
II	Moderate	FEV_1/FVC <0.7 and FEV_1 ≥50% but <80% predicted
III	Severe	FEV_1/FVC <0.7 and FEV_1 ≥30% but <50% predicted
IV	Very severe	FEV_1/FVC <0.7 and FEV_1 <30% predicted

Abbreviations: COPD, chronic obstructive pulmonary disease; FEV1, forced expiratory volume in 1 s; FVC, forced vital capacity; GOLD, Global Initiative for Chronic Obstructive Lung Disease.

Source: Reproduced with permission from the Global Strategy for Diagnosis, Management and Prevention of COPD 2021, ©.

destruction, and the development of emphysema are present in most persons with COPD; however, they appear to be relatively independent processes, and their relative contributions to obstruction vary from one person to another. The early stages of COPD, based on the severity of airflow obstruction (Table 292-1), appear to be primarily associated with medium and small airway disease with the majority of Global Initiative for Chronic Obstructive Lung Disease (GOLD) spirometric airflow obstruction stage 1 and stage 2 subjects demonstrating little or no emphysema. The early development of chronic airflow obstruction is driven by small airway disease. Advanced stages of COPD (GOLD stages 3 and 4) are typically characterized by extensive emphysema, although there are a small number of subjects with very severe (GOLD stage 4) obstruction with virtually no emphysema. The subjects at greatest risk of progression in COPD are those with both aggressive airway disease and emphysema. Thus, finding emphysema (by chest computed tomography [CT]) either early or late in the disease process suggests enhanced risk for disease progression.

LARGE AIRWAYS

Cigarette smoking often results in mucus gland enlargement and goblet cell hyperplasia, leading to cough and mucus production that define chronic bronchitis, but these abnormalities are not directly related to airflow obstruction. In response to cigarette smoking, goblet cells increase not only in number but also in extent through the bronchial tree. Bronchi also undergo squamous metaplasia, predisposing to carcinogenesis and disrupting mucociliary clearance. Although not as prominent as in asthma, patients may have smooth-muscle hypertrophy and bronchial hyperreactivity leading to airflow obstruction. Neutrophil influx has been associated with purulent sputum during respiratory tract infections. Independent of its proteolytic activity, neutrophil elastase is among the most potent secretagogues identified.

SMALL AIRWAYS

The major site of increased resistance in most individuals with COPD is in airways ≤2 mm diameter. Characteristic cellular changes include goblet cell metaplasia, with these mucus-secreting cells replacing surfactant-secreting Club cells. Smooth-muscle hypertrophy may also be present. Luminal narrowing can occur by fibrosis, excess mucus, edema, and cellular infiltration. Reduced surfactant may increase surface tension at the air-tissue interface, predisposing to airway narrowing or collapse. Respiratory bronchiolitis with mononuclear inflammatory cells collecting in distal airway tissues may cause proteolytic destruction of elastic fibers in the respiratory bronchioles and alveolar ducts where the fibers are concentrated as rings around alveolar entrances. Narrowing and drop-out of small airways precede the onset of emphysematous destruction. Advanced COPD has been shown to be associated with a loss of many of the smaller airways and a similar significant loss of the lung microvasculature.

LUNG PARENCHYMA

Emphysema is characterized by destruction of gas-exchanging air spaces, i.e., the respiratory bronchioles, alveolar ducts, and alveoli. Large numbers of macrophages accumulate in respiratory bronchioles of essentially all smokers. Neutrophils, B lymphocytes, and T lymphocytes, particularly CD8+ cells, are also increased in the alveolar space of smokers. Alveolar walls become perforated and later obliterated with coalescence of the delicate alveolar structure into large emphysematous air spaces.

Emphysema is classified into distinct pathologic types, which include centrilobular, panlobular, and paraseptal (Fig 292-2). *Centrilobular emphysema*, the type most frequently associated with cigarette smoking, is characterized by enlarged air spaces found (initially) in association with respiratory bronchioles. Centrilobular emphysema is usually most prominent in the upper lobes and superior segments of lower lobes and is often quite focal. *Panlobular emphysema* refers to abnormally large air spaces evenly distributed within and across acinar units. Panlobular emphysema is commonly observed in patients with α_1AT deficiency, which has a predilection for the lower lobes. Paraseptal emphysema occurs in 10–15% of cases and is distributed along the pleural margins with relative sparing of the lung core or central regions. It is commonly associated with significant airway inflammation and with centrilobular emphysema.

PATHOPHYSIOLOGY

Persistent reduction in forced expiratory flow rates is the classic definition of COPD. Hyperinflation with increases in the residual volume and the residual volume/total lung capacity ratio, nonuniform distribution of ventilation, and ventilation-perfusion mismatching also occur.

AIRFLOW OBSTRUCTION

Airflow obstruction, also known as airflow limitation, is typically determined for clinical purposes by spirometry, which involves maximal forced expiratory maneuvers after the subject has inhaled to total lung capacity. Key parameters obtained from spirometry include the volume of air exhaled within the first second of the forced expiratory maneuver (FEV_1) and the total volume of air exhaled during the entire spirometric maneuver (forced vital capacity [FVC]). Patients with airflow obstruction related to COPD have a chronically reduced ratio of FEV_1/FVC. In contrast to asthma, the reduced FEV_1 in COPD seldom shows large improvements to inhaled bronchodilators, although improvements up to 15% are common.

HYPERINFLATION

Lung volumes are also routinely assessed in pulmonary function testing. In COPD, there is often "air trapping" (increased residual volume and increased ratio of residual volume to total lung capacity)

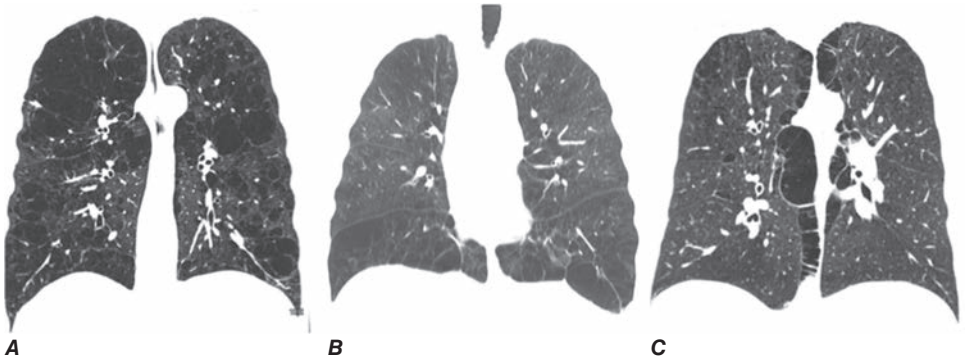


FIGURE 292-2 CT patterns of emphysema. ***A.*** Centrilobular emphysema with severe upper lobe involvement in a 68-year-old man with a 70-pack-year smoking history but forced expiratory volume in 1 s (FEV_1) 81% predicted (GOLD spirometry grade 1). ***B.*** Panlobular emphysema with diffuse loss of lung parenchymal detail predominantly in the lower lobes in a 64-year-old man with severe α_1 antitrypsin (α_1AT) deficiency. ***C.*** Paraseptal emphysema with marked airway inflammation in a 52-year-old woman with a 37-pack-year smoking history and FEV_1 40% predicted.

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and progressive hyperinflation (increased total lung capacity) in more advanced disease. Hyperinflation of the thorax during tidal breathing preserves maximum expiratory airflow, because as lung volume increases, elastic recoil pressure increases, and airways enlarge so that airway resistance decreases.

Despite compensating for airway obstruction, hyperinflation can push the diaphragm into a flattened position with a number of adverse effects. First, by decreasing the zone of apposition between the diaphragm and the abdominal wall, positive abdominal pressure during inspiration is not applied as effectively to the chest wall, hindering rib cage movement and impairing inspiration. Second, because the muscle fibers of the flattened diaphragm are shorter than those of a more normally curved diaphragm, they are less capable of generating inspiratory pressures than normal. Third, the flattened diaphragm must generate greater tension to develop the transpulmonary pressure required to produce tidal breathing. Fourth, the thoracic cage is distended beyond its normal resting volume, and during tidal breathing, the inspiratory muscles must do work to overcome the resistance of the thoracic cage to further inflation instead of gaining the normal assistance from the chest wall recoiling outward toward its resting volume.

GAS EXCHANGE

Although there is considerable variability in the relationships between the FEV_1 and other physiologic abnormalities in COPD, certain generalizations may be made. The partial pressure of oxygen in arterial blood Pao_2 usually remains near normal until the FEV_1 is decreased to below 50% of predicted, and even much lower FEV_1 values can be associated with a normal Pao_2, at least at rest. An elevation of arterial level of carbon dioxide ($Paco_2$) is not expected until the FEV_1 is <25% of predicted and even then may not occur. Pulmonary arterial hypertension severe enough to cause cor pulmonale and right ventricular failure due to COPD typically occurs in individuals who have marked decreases in FEV_1 (<25% of predicted) and chronic hypoxemia (Pao_2 <55 mmHg); however, some patients develop significant pulmonary arterial hypertension independent of COPD severity (Chap. 283).

Nonuniform ventilation and ventilation-perfusion mismatching are characteristic of COPD, reflecting the heterogeneous nature of the disease process within the airways and lung parenchyma. Physiologic studies are consistent with multiple parenchymal compartments having different rates of ventilation due to regional differences in compliance and airway resistance. Ventilation-perfusion mismatching accounts for essentially all of the reduction in Pao_2 that occurs in COPD; shunting is minimal. This finding explains the effectiveness of modest elevations of inspired oxygen in treating hypoxemia due to COPD and therefore the need to consider problems other than COPD when hypoxemia is difficult to correct with modest levels of supplemental oxygen.

RISK FACTORS

CIGARETTE SMOKING

By 1964, the Advisory Committee to the Surgeon General of the United States had concluded that cigarette smoking was a major risk factor for mortality from chronic bronchitis and emphysema. Subsequent longitudinal studies have shown accelerated decline in FEV_1 in a dose-response relationship to the intensity of cigarette smoking, which is typically expressed as pack-years (average number of packs of cigarettes smoked per day multiplied by the total number of years of smoking). This dose-response relationship between reduced pulmonary function and cigarette smoking intensity accounts, at least in part, for the higher prevalence rates of COPD with increasing age. The historically higher rate of smoking among males is the likely explanation for the higher prevalence of COPD among males; however, the prevalence of COPD among females is increasing as the gender gap in smoking rates has diminished in the past 50 years.

Although the causal relationship between cigarette smoking and the development of COPD has been absolutely proved, there is considerable individual variability in the response to smoking. Pack-years of cigarette smoking is the most highly significant predictor of FEV_1 (Fig. 292-3), but only 15% of the variability in FEV_1 is explained by

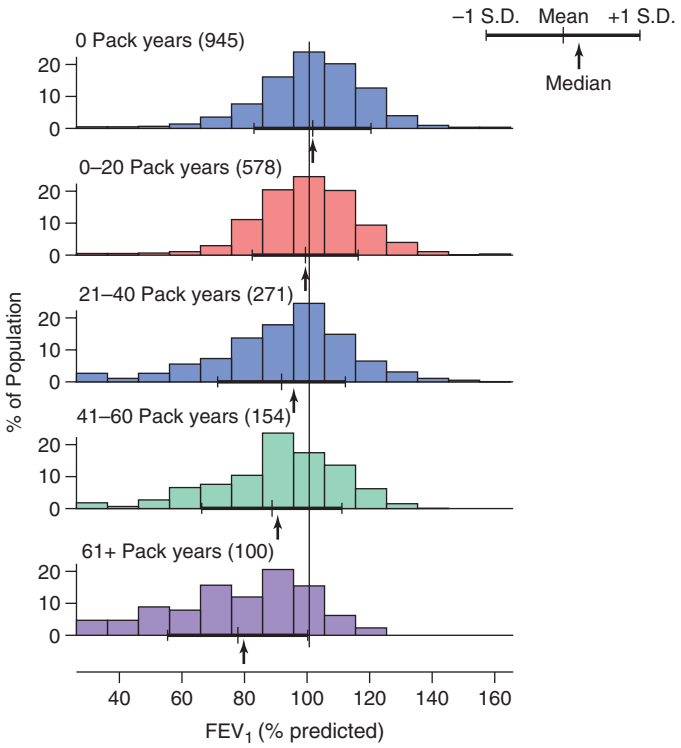


FIGURE 292-3 Distributions of forced expiratory volume in 1 s (FEV_1) values in a general population sample, stratified by pack-years of smoking. Means, medians, and ±1 standard deviation of percent predicted FEV_1 are shown for each smoking group. Although a dose-response relationship between smoking intensity and FEV_1 was found, marked variability in pulmonary function was observed among subjects with similar smoking histories. S.D., standard deviation. *(Reproduced with permission from B Burrows: Quantitative relationships between cigarette smoking and ventilatory function. Am Rev Respir Dis 115:195, 1997.)*

pack-years. This finding suggests that additional environmental and/or genetic factors contribute to the impact of smoking on the development of chronic airflow obstruction. Nonetheless, many patients with a history of cigarette smoking with normal spirometry have evidence for worse health-related quality of life, reduced exercise capacity, and emphysema and/or airway disease on chest CT evaluation; thus, they have not escaped the harmful effects of cigarette smoking. While they do not meet the classic definition of COPD based on population normals for FEV_1 and FEV_1/FVC, studies have shown that these subjects overall have a shift toward lower FEV_1 values, which is consistent with obstruction on an individual level.

Although cigar and pipe smoking may also be associated with the development of COPD, the evidence supporting such associations is less compelling, likely related to the lower dose of inhaled tobacco by-products during cigar and pipe smoking. The impact of electronic cigarettes and vaping on the development and progression of COPD has not yet been determined.

AIRWAY RESPONSIVENESS AND COPD

A tendency for increased bronchoconstriction in response to a variety of exogenous stimuli, including methacholine and histamine, is one of the defining features of asthma (Chap. 287). However, many patients with COPD also share this feature of airway hyperresponsiveness. In older subjects, there is considerable overlap between persons with a history of chronic asthma and smokers with COPD in terms of airway responsiveness, airflow obstruction, and pulmonary symptoms. The origin of asthma is viewed in many patients as an allergic disease while COPD is thought to primarily result from smoking-related inflammation and damage; however, they likely share common environmental and genetic factors and the chronic form in older subjects can present similarly. This is particularly relevant for childhood asthmatic subjects who become chronic smokers.

Longitudinal studies that compared airway responsiveness to subsequent decline in pulmonary function have demonstrated that increased

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airway responsiveness is clearly a significant predictor of subsequent decline in pulmonary function. A study from the Childhood Asthma Management Program identified four lung function trajectories in children with persistent asthma. Asthmatics with reduced lung function early in life were more likely to meet spirometric criteria for COPD in early adulthood. Both asthma and airway hyperresponsiveness are risk factors for COPD.

RESPIRATORY INFECTIONS

The impact of adult respiratory infections on decline in pulmonary function is controversial, but significant long-term reductions in pulmonary function are not typically seen following an individual episode of acute bronchitis or pneumonia. However, respiratory infections are important causes of COPD exacerbations, and recent results from the COPDGene and ECLIPSE studies suggest that COPD exacerbations are associated with increased loss of lung function longitudinally, particularly among those individuals with better baseline lung function levels. The impact of the effects of childhood respiratory illnesses on the subsequent development of COPD has been difficult to assess due to a lack of adequate longitudinal data, but recent studies have suggested that childhood pneumonia may lead to increased risk for COPD later in life.

OCCUPATIONAL EXPOSURES

Increased respiratory symptoms and airflow obstruction have been suggested to result from exposure to dust and fumes. Several specific occupational exposures, including coal mining, gold mining, and cotton textile dust, have been implicated as risk factors for chronic airflow obstruction. Although nonsmokers in these occupations can develop some reductions in FEV_1, the importance of dust exposure as a risk factor for COPD, independent of cigarette smoking, is not certain for most of these exposures. However, among coal miners, coal mine dust exposure was a significant risk factor for emphysema in both smokers and nonsmokers. In most cases, the magnitude of these occupational exposures on COPD risk is likely substantially less important than the effect of cigarette smoking.

AMBIENT AIR POLLUTION

Some investigators have reported increased respiratory symptoms in those living in urban compared to rural areas, which may relate to increased pollution in the urban settings. However, the relationship of air pollution to chronic airflow obstruction remains unproved. Prolonged exposure to smoke produced by biomass combustion—a common mode of cooking in some countries—also appears to be a significant risk factor for COPD, particularly among women.

PASSIVE, OR SECOND-HAND, SMOKING EXPOSURE

Exposure of children to maternal smoking results in significantly reduced lung growth. In utero, tobacco smoke exposure also contributes to significant reductions in postnatal pulmonary function. Although passive smoke exposure has been associated with reductions in pulmonary function, the importance of this risk factor in the development of the severe pulmonary function reductions often observed in COPD remains uncertain.

GENETIC CONSIDERATIONS



Although cigarette smoking is the major environmental risk factor for the development of COPD, the development of airflow obstruction in smokers is highly variable. Severe α_1AT deficiency is a proven genetic risk factor for COPD; there is increasing evidence that other genetic determinants also exist.

α_1 Antitrypsin Deficiency Many variants of the protease inhibitor (PI or *SERPINA1*) locus that encodes α_1AT have been described. The common M allele is associated with normal α_1AT levels. The S allele, associated with slightly reduced α_1AT levels, and the Z allele, associated with markedly reduced α_1AT levels, also occur with frequencies of >1% in most white populations. Rare individuals inherit null alleles, which lead to the absence of any α_1AT production through a heterogeneous collection of mutations. Individuals with two Z alleles or one Z and one null allele are referred to as Pi^Z, which is the most common form of severe α_1AT deficiency.

Although only ~1% of COPD patients are found to have severe α_1AT deficiency as a contributing cause of COPD, these patients demonstrate that genetic factors can have a profound influence on the susceptibility for developing COPD. Pi^Z individuals often develop early-onset COPD, but the ascertainment bias in the published series of Pi^Z individuals—which have usually included many Pi^Z subjects who were tested for α_1AT deficiency because they had COPD—means that the fraction of Pi^Z individuals who will develop COPD and the age-of-onset distribution for the development of COPD in Pi^Z subjects remain unknown. Approximately 1 in 3000 individuals in the United States inherits severe α_1AT deficiency, but only a small minority of these individuals has been identified. The clinical laboratory test used most frequently to screen for α_1AT deficiency is measurement of the immunologic level of α_1AT in serum (see "Laboratory Findings").

A significant percentage of the variability in pulmonary function among Pi^Z individuals is explained by cigarette smoking; cigarette smokers with severe α_1AT deficiency are more likely to develop COPD at early ages. However, the development of COPD in Pi^Z subjects, even among current or ex-smokers, is not absolute. Among Pi^Z nonsmokers, impressive variability has been noted in the development of airflow obstruction. Asthma and male gender also appear to increase the risk of COPD in Pi^Z subjects. Other genetic and/or environmental factors likely contribute to this variability.

Specific treatment in the form of α_1AT augmentation therapy is available for severe α_1AT deficiency as a weekly IV infusion (see "Treatment," below).

The risk of lung disease in heterozygous Pi^{MZ} individuals, who have intermediate serum levels of α_1AT (~60% of Pi^{MM} levels), has been controversial. Several recent studies have demonstrated that Pi^{MZ} subjects who smoke are likely at increased risk for the development of COPD. However, α_1AT augmentation therapy is not recommended for use in Pi^{MZ} subjects.

Other Genetic Risk Factors Studies of pulmonary function measurements performed in general population samples have indicated that genetic factors other than PI type influence variation in pulmonary function. Familial aggregation of airflow obstruction within families of COPD patients has also been demonstrated.

GWAS have identified >80 regions of the genome that contain COPD susceptibility loci, including a region near the *HHIP* gene on chromosome 4, a cluster of genes on chromosome 15 (including components of the nicotinic acetylcholine receptor and another gene, *IREB2*, related to mitochondrial iron regulation), and a region within a gene of unknown function (*FAM13A*). As with most other complex diseases, the risk associated with individual GWAS loci is modest, but these genetic determinants may identify important biological pathways related to COPD. Gene-targeted murine models for *HHIP*, *FAM13A*, and *IREB2* exposed to chronic cigarette smoke had altered emphysema susceptibility, suggesting that those genes are likely to be involved in COPD pathogenesis.

NATURAL HISTORY

The effects of cigarette smoking on pulmonary function appear to depend on the intensity of smoking exposure, the timing of smoking exposure during growth and development, and the baseline lung function of the individual; other environmental factors may have similar effects. Most individuals follow a steady trajectory of increasing pulmonary function with growth during childhood and adolescence, followed by a plateau in early adulthood, and then gradual decline with aging. Individuals appear to track in their quantile of pulmonary function based on environmental and genetic factors that put them on different tracks. The risk of eventual mortality from COPD is closely associated with reduced levels of FEV_1. A graphic depiction of the natural history of COPD is shown as a function of the influences on tracking curves of FEV_1 in **Fig. 292-4**. Death or disability from COPD can result from a normal rate of decline after a reduced growth

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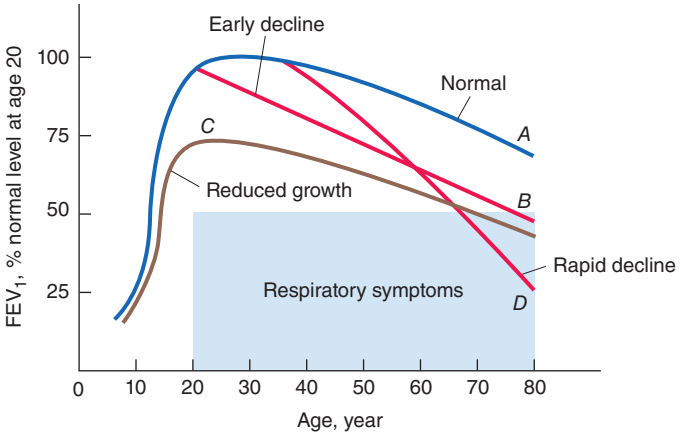


FIGURE 292-4 Hypothetical tracking curves of forced expiratory volume in 1 s (FEV_1) for individuals throughout their life spans. The normal pattern of growth and decline with age is shown by curve *A*. Significantly reduced FEV_1 (<65% of predicted value at age 20) can develop from a normal rate of decline after a reduced pulmonary function growth phase (curve *C*), early initiation of pulmonary function decline after normal growth (curve *B*), or accelerated decline after normal growth (curve *D*). *(From B Rijcken: Doctoral dissertation, p 133, University of Groningen, 1991; with permission.)*

phase (curve *C*), an early initiation of pulmonary function decline after normal growth (curve *B*), or an accelerated decline after normal growth (curve *D*). Although accelerated rates of lung function decline have classically been associated with COPD, recent analyses of several population-based cohorts demonstrated that many subjects meeting the spirometric criteria for COPD had reduced growth but normal rates of lung function decline. The rate of decline in pulmonary function can be modified by changing environmental exposures (i.e., quitting smoking), with smoking cessation at an earlier age providing a more beneficial effect than smoking cessation after marked reductions in pulmonary function have already developed. The absolute annual loss in FEV_1 tends to be highest in mild COPD and lowest in very severe COPD. Multiple genetic factors influence the level of pulmonary function achieved during growth.

In chronic smokers, substantial chest CT changes (emphysema and airway wall thickening) have been identified in subjects with normal physiology (normal FEV_1 and FEV_1/FVC). COPD in these subjects commonly progresses in two primary patterns. Subjects with an emphysema-predominant pattern show emphysema early and classically progress through GOLD 1 to GOLD 2–4. Subjects with an airway disease–predominant pattern typically show initial evidence of airway inflammation and progress with little emphysema early as FEV_1 falls while retaining a normal FEV_1/FVC ratio. This is termed *preserved ratio–impaired spirometry* (PRISm) physiology. These subjects tend to develop emphysema late and can progress directly to GOLD 3 and 4 with severe, end-stage COPD.

CLINICAL PRESENTATION

HISTORY

The three most common symptoms in COPD are cough, sputum production, and exertional dyspnea. Many patients have such symptoms for months or years before seeking medical attention. Although the development of airflow obstruction is a gradual process, many patients date the onset of their disease to an acute illness or exacerbation. A careful history, however, usually reveals the presence of respiratory symptoms prior to the acute exacerbation. The development of exertional dyspnea, often described as increased effort to breathe, heaviness, air hunger, or gasping, can be insidious. It is best elicited by a careful history focused on typical physical activities and how the patient's ability to perform them has changed. Activities involving significant arm work, particularly at or above shoulder level, are particularly difficult for many patients with COPD. Conversely, activities that allow the patient to brace the arms and use accessory muscles of respiration are better tolerated. Examples of such activities include pushing a shopping cart or walking on a treadmill. As COPD advances, the principal feature is worsening dyspnea on exertion with increasing intrusion on the ability to perform vocational or avocational activities. In the most advanced stages, patients are breathless doing basic activities of daily living.

Accompanying worsening airflow obstruction is an increased frequency of exacerbations (described below). Patients may also develop resting hypoxemia and require institution of supplemental oxygen.

PHYSICAL FINDINGS

In the early stages of COPD, patients usually have an entirely normal physical examination. Current smokers may have signs of active smoking, including an odor of smoke or nicotine staining of fingernails. In patients with more severe disease, the physical examination of the lungs is notable for a prolonged expiratory phase and may include expiratory wheezing. In addition, signs of hyperinflation include a barrel chest and enlarged lung volumes with poor diaphragmatic excursion as assessed by percussion. Patients with severe airflow obstruction may also exhibit use of accessory muscles of respiration, sitting in the characteristic "tripod" position to facilitate the actions of the sternocleidomastoid, scalene, and intercostal muscles. Patients may develop cyanosis, visible in the lips and nail beds.

Traditional teaching is that patients with predominant emphysema, termed "pink puffers," are thin, noncyanotic at rest, and have prominent use of accessory muscles, and patients with chronic bronchitis are more likely to be heavy and cyanotic ("blue bloaters"). However, current evidence demonstrates that most patients have elements of both chronic bronchitis and emphysema and that the physical examination does not reliably differentiate the two entities.

Advanced disease may be accompanied by cachexia, with significant weight loss and diffuse loss of subcutaneous adipose tissue. This syndrome has been associated with both inadequate oral intake and elevated levels of inflammatory cytokines (TNF-α). Such wasting is an independent poor prognostic factor in COPD. Some patients with advanced disease have paradoxical inward movement of the rib cage with inspiration (Hoover's sign), the result of alteration of the vector of diaphragmatic contraction on the rib cage due to chronic hyperinflation.

Signs of overt right heart failure, termed *cor pulmonale*, are relatively infrequent since the advent of supplemental oxygen therapy.

Clubbing of the digits is not a sign of COPD, and its presence should alert the clinician to initiate an investigation for causes of clubbing. In COPD patients, the development of lung cancer is the most likely explanation for newly developed clubbing.

LABORATORY FINDINGS

The hallmark of COPD is airflow obstruction (discussed above). Pulmonary function testing shows airflow obstruction with a reduction in FEV_1 and FEV_1/FVC **(Chap. 285)**. With worsening disease severity, lung volumes may increase, resulting in an increase in total lung capacity, functional residual capacity, and residual volume. In patients with emphysema, the diffusing capacity may be reduced, reflecting the lung parenchymal destruction characteristic of the disease. The degree of airflow obstruction is an important prognostic factor in COPD and is the basis for the GOLD spirometric severity classification (Table 292-1). Although the degree of airflow obstruction *generally* correlates with the presence and severity of respiratory symptoms, exacerbations, emphysema, and hypoxemia, the correlations are far from perfect. Thus, clinical features should be carefully assessed in each individual patient with COPD to determine the most appropriate therapies. It has been shown that a multifactorial index (BODE), incorporating airflow obstruction, exercise performance, dyspnea, and body mass index, is a better predictor of mortality. Recently, GOLD added additional elements to their COPD classification system incorporating respiratory symptoms and exacerbation history; these metrics are used to guide COPD treatment (see below).

Arterial blood gases and oximetry may demonstrate resting or exertional hypoxemia. Arterial blood gases provide additional information about alveolar ventilation and acid-base status by measuring arterial

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P_{CO_2} and pH. The change in pH with P_{CO_2} is 0.08 units/10 mmHg acutely and 0.03 units/10 mmHg in the chronic state. Knowledge of the arterial pH therefore allows the classification of ventilatory failure, defined as P_{CO_2} >45 mmHg, into acute or chronic conditions with acute respiratory failure being associated with acidemia. The arterial blood gas is an important component of the evaluation of patients presenting with symptoms of an exacerbation. An elevated hematocrit suggests the presence of chronic hypoxemia, as does the presence of signs of right ventricular hypertrophy.

Radiographic studies may assist in the classification of the type of COPD. Increased lung volumes and flattening of the diaphragm suggest hyperinflation but do not provide information about chronicity of the changes. Obvious bullae, paucity of parenchymal markings, or hyperlucency on chest x-ray suggests the presence of emphysema. Chest CT scan is the current definitive test for establishing the presence or absence of emphysema, the pattern of emphysema, and the presence of significant disease involving medium and large airways (Fig. 292-2). It also enables the discovery of coexisting interstitial lung disease and bronchiectasis. Smokers with COPD are at high risk for development of lung cancer, which can be identified on a chest CT scan. In advanced COPD, CT scans can help determine the possible value of surgical therapy (described below).

Recent guidelines have suggested testing for α_1AT deficiency in all subjects with COPD or asthma with chronic airflow obstruction. Measurement of the serum α_1AT level is a reasonable initial test. For subjects with low α_1AT levels, the definitive diagnosis of α_1AT deficiency requires PI type determination. This is typically performed by isoelectric focusing of serum or plasma, which reflects the genotype at the PI locus for the common alleles and many of the rare PI alleles as well. Molecular genotyping can be performed for the common PI alleles (M, S, and Z), and DNA sequencing can detect other rare deficiency variants.

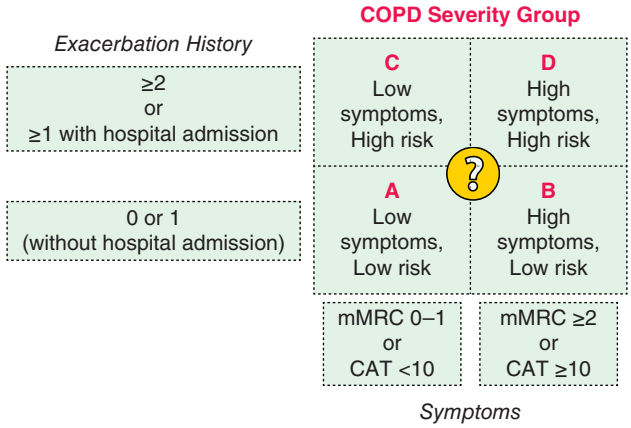


FIGURE 292-5 Chronic obstructive pulmonary disease (COPD) severity assessment. COPD severity categories are based on respiratory symptoms (based on the Modified Medical Research Council Dyspnea Scale [mMRC] or COPD Assessment Test [CAT]) and annual frequency of COPD exacerbations. The mMRC provides a single number for degree of breathlessness: 0—only with strenuous activity; 1—hurrying on level ground or walking up a slight hill; 2—walk slower than peers or stop walking at their own pace; 3—walking about 100 yards or after a few minutes on level ground; 4—too breathless to leave the house or when dressing. The CAT is an eight-item COPD health status measure with Likert scale responses for questions about cough, phlegm, chest tightness, dyspnea on one flight of stairs, limitation in home activities, confidence in leaving the home, sleep, and energy. Range of total score is 0–40. Both mMRC and CAT are available from Global Strategy for the Diagnosis, Management and Prevention of COPD, Global Initiative for Chronic Obstructive Lung Disease (GOLD) 2017. *(Reproduced with permission from Global Strategy for Diagnosis, Management and Prevention of COPD 2017, ©.)*

TREATMENT

Chronic Obstructive Pulmonary Disease

STABLE PHASE COPD

The two main goals of therapy are to provide symptomatic relief (reduce respiratory symptoms, improve exercise tolerance, and improve health status) and reduce future risk (prevent disease progression, prevent and treat exacerbations, and reduce mortality). The institution of therapies should be based on symptom assessment, benefits of therapy, potential risks, and costs. **Figure 292-5** provides the currently suggested categories of COPD patients based on respiratory symptoms and risk for exacerbations. Response to therapy should be assessed, and decisions should be made whether or not to continue or alter treatment.

Three interventions—smoking cessation, oxygen therapy in chronically hypoxemic patients, and lung volume reduction surgery (LVRS) in selected patients with emphysema—have been demonstrated to improve survival of patients with COPD. Recent studies indicate that triple inhaled therapy (long-acting beta agonist bronchodilator, long-acting muscarinic antagonist bronchodilator and inhaled corticosteroid) reduces mortality in selected patients with COPD. There is a suggestion that inhaled LAMA bronchodilators may reduce mortality.

PHARMACOTHERAPY

Smoking Cessation (See also Chap. 454) It has been shown that middle-aged smokers who were able to successfully stop smoking experienced a significant improvement in the rate of decline in pulmonary function, often returning to annual changes similar to that of nonsmoking patients. In addition, smoking cessation improves survival. Thus, all patients with COPD should be strongly urged to quit smoking and educated about the benefits of quitting. An emerging body of evidence demonstrates that combining pharmacotherapy with traditional supportive approaches considerably enhances the chances of successful smoking cessation. There are three principal pharmacologic approaches to the problem: nicotine replacement therapy available as gum, transdermal patch, lozenge, inhaler, and nasal spray; bupropion; and varenicline, a nicotinic acid receptor agonist/antagonist. Current recommendations from the U.S. Surgeon General are that all adult, nonpregnant smokers considering quitting be offered pharmacotherapy, in the absence of any contraindication to treatment. Smoking cessation counseling is also recommended and free counseling is available through state Smoking QuitLines.

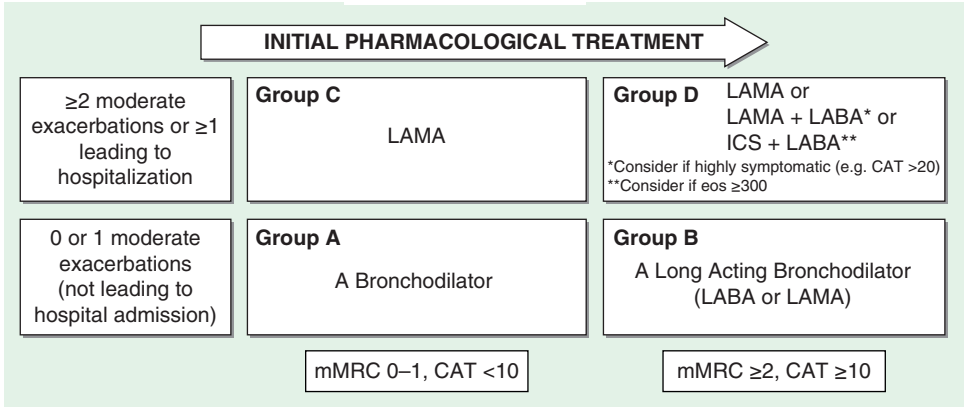
Bronchodilators In general, bronchodilators are the primary treatment for almost all patients with COPD and are used for symptomatic benefit and to reduce exacerbations. The inhaled route is preferred for medication delivery, because side effects are less than with systemic medication delivery. In symptomatic patients, both regularly scheduled use of long-acting agents and as-needed short-acting medications are indicated. **Figure 292-6** provides suggestions for prescribing inhaled medication therapy based on grouping patients by severity of symptoms and risk of exacerbations.

Muscarinic Antagonists Short-acting ipratropium bromide improves symptoms with acute improvement in FEV_1. Long-acting muscarinic antagonists (LAMA, including aclidinium, glycopyrrolate, glycopyrronium, revefenacin, tiotropium, and umeclidinium) improve symptoms and reduce exacerbations. In a large randomized clinical trial, there was a trend toward reduced mortality rate in tiotropium-treated patients that approached statistical significance. Side effects are minor; dry mouth is the most frequent side effect.

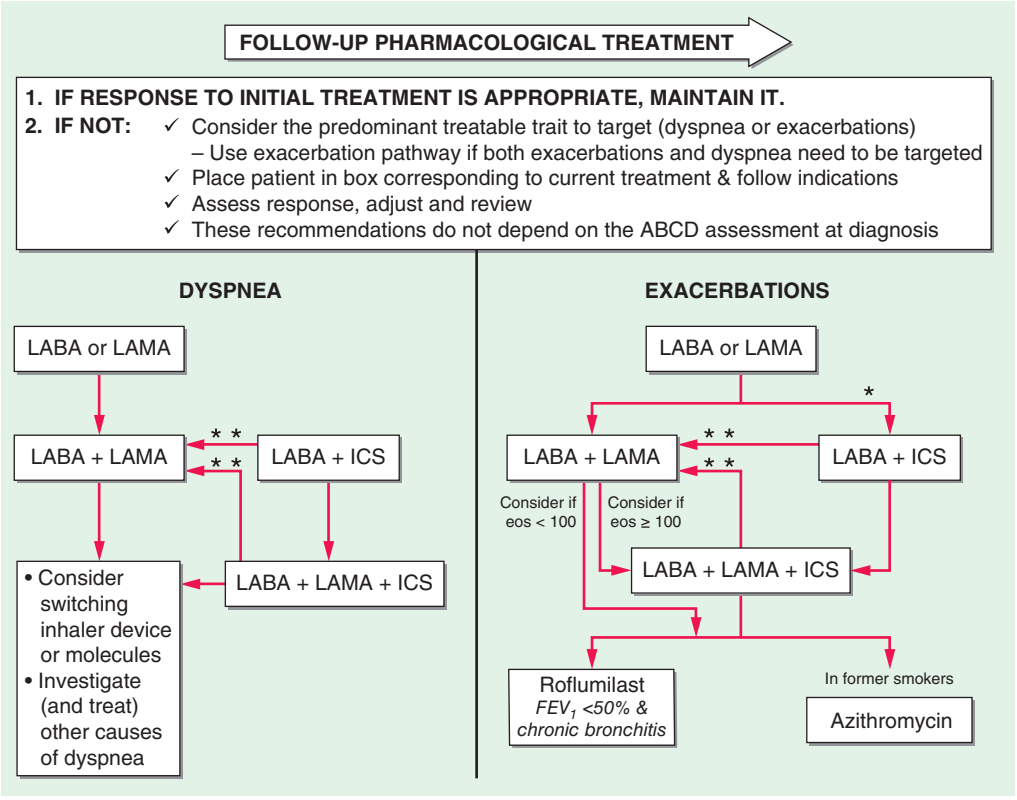
Beta Agonists Short-acting beta agonists ease symptoms with acute improvements in lung function. Long-acting beta agonists (LABAs) provide symptomatic benefit and reduce exacerbations, though to a lesser extent than an LAMA. Currently available long-acting inhaled beta agonists are arformoterol, formoterol, indacaterol, olodaterol, salmeterol, and vilanterol. The main side effects are tremor and tachycardia.

Combinations of Beta Agonist–Muscarinic Antagonist The combination inhaled long-acting beta agonist and muscarinic antagonist therapy has been demonstrated to provide improvement in

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A



B

FIGURE 292-6 Medication therapy for stable chronic obstructive pulmonary disease (COPD). Initial pharmacological therapy (Panel A) is based on both COPD exacerbations and respiratory symptoms (assessed through the modified Medical Research Council (mMRC) dyspnea questionnaire or the COPD Assessment Test (CAT)). Follow-up pharmacological therapy (Panel B) is based on response to treatment initiation and reassessment of symptoms and exacerbations. Global Strategy for the Diagnosis, Management and Prevention of COPD, Global Initiative for Chronic Obstructive Lung Disease (GOLD), 2021. *For Panel B: consider if eos ≥300 or eos ≥100 AND ≥2 moderate exacerbations/1 hospitalization. **Consider de-escalation of ICS or switch if pneumonia, inappropriate original indication or lack of response to ICS. CAT™, COPD Assessment Test™; Eos, blood eosinophil count in cells per microliter; FEV1, forced expiratory volume in 1 second; ICS, inhaled corticosteroid; LABA, long-acting beta agonist; LAMA, long-acting muscarinic antagonist; mMRC, modified Medical Research Council dyspnea questionnaire. *(Reproduced with permission from the Global Strategy for Diagnosis, Management and Prevention of COPD 2021, ©.)*

lung function that is greater than either agent alone and reduces exacerbations.

Inhaled Corticosteroids The main role of ICS is to reduce exacerbations. In population studies, patients with an eosinophil count of <100 cells per microliter do not benefit, while benefit increases as eosinophil counts rise above 100. ICS are never used alone in COPD due to little symptomatic benefit but, rather, are combined with a LABA or used with a LABA and LAMA. Their use has been associated with increased rates of oropharyngeal candidiasis and pneumonia and in some studies an increased rate of loss of bone density and development of cataracts. A trial of ICS should be considered in patients with frequent exacerbations, defined as two or more per year or in patients hospitalized with one exacerbation. In stable patients without exacerbations, ICS withdrawal may be considered. Patients who continue to smoke cigarettes do not benefit as greatly from ICS use. Although ICS withdrawal does not lead to an increase in exacerbations, there may be a small decline in lung function.

Oral Glucocorticoids The chronic use of oral glucocorticoids for treatment of COPD is not recommended because of an unfavorable benefit/risk ratio. The chronic use of oral glucocorticoids is associated with significant side effects, including osteoporosis, weight

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gain, cataracts, glucose intolerance, and increased risk of infection. A study demonstrated that patients tapered off chronic low-dose prednisone (~10 mg/d) did not experience any adverse effect on the frequency of exacerbations, health-related quality of life, or lung function.

Theophylline Theophylline produces modest improvements in airflow and vital capacity, but is not first-line therapy due to side effects and drug interactions. Nausea is a common side effect; tachycardia and tremor have also been reported. Monitoring of blood theophylline levels is required to minimize toxicity.

PDE4 Inhibitors The selective phosphodiesterase 4 (PDE4) inhibitor roflumilast has been demonstrated to reduce exacerbation frequency in patients with severe COPD, chronic bronchitis, and a prior history of exacerbations; its effects on airflow obstruction and symptoms are modest, and side effects (including nausea, diarrhea, and weight loss) are common.

Antibiotics There are strong data implicating bacterial infection as a precipitant of a substantial portion of exacerbations. A randomized clinical trial of azithromycin, chosen for both its anti-inflammatory and antimicrobial properties, administered daily to subjects with a history of exacerbation in the past 6 months demonstrated a reduced exacerbation frequency and longer time to first exacerbation in the macrolide-treated cohort (hazard ratio, 0.73). Azithromycin was most effective in older patients and milder GOLD stages; there was little benefit in current smokers.

Oxygen Supplemental O_2 is the only pharmacologic therapy demonstrated to unequivocally decrease mortality in patients with COPD. For patients with resting hypoxemia (resting O_2 saturation ≤88% in any patient or ≤89% with signs of pulmonary arterial hypertension, right heart failure or erythrocytosis), the use of O_2 has been demonstrated to have a significant impact on mortality. Patients meeting these criteria should be on continuous oxygen supplementation because the mortality benefit is proportional to the number of hours per day oxygen is used. Various delivery systems are available, including portable systems that patients may carry to allow mobility outside the home.

A recent study failed to demonstrate mortality benefits to COPD patients with moderate hypoxemia at rest or with hypoxemia only with activity.

α_1AT Augmentation Therapy Specific treatment in the form of IV α_1AT augmentation therapy is available for individuals with severe α_1AT deficiency. Despite sterilization procedures for these blood-derived products and the absence of reported cases of viral infection from therapy, some physicians recommend hepatitis B vaccination prior to starting augmentation therapy. Although biochemical efficacy of α_1AT augmentation therapy has been shown, the benefits of α_1AT augmentation therapy are controversial. A randomized study suggested a reduction in emphysema progression in patients receiving α_1AT augmentation therapy. Eligibility for α_1AT augmentation therapy requires a serum α_1AT level <11 μ*M* (~50 mg/dL). Typically, Pi^Z individuals will qualify, although other rare types associated with severe deficiency (e.g., null-null) are also eligible. Because only a fraction of individuals with severe α_1AT deficiency will develop COPD, α_1AT augmentation therapy is not recommended for severely α_1AT-deficient persons with normal pulmonary function and a normal chest CT scan.

NONPHARMACOLOGIC THERAPIES

Patients with COPD should receive the influenza vaccine annually. Pneumococcal vaccines and vaccination for *Bordetella pertussis* are recommended.

Pulmonary Rehabilitation This refers to a comprehensive treatment program that incorporates exercise, education, and psychosocial and nutritional counseling. In COPD, pulmonary rehabilitation has been demonstrated to improve health-related quality of life, dyspnea, and exercise capacity. It has also been shown to reduce rates of hospitalization over a 6- to 12-month period.

Lung Volume Reduction Surgery In carefully selected patients with emphysema, surgery to remove the most emphysematous portions of lung improves exercise capacity, lung function, and survival. The anatomic distribution of emphysema and postrehabilitation exercise capacity are important prognostic characteristics. Patients with upper lobe–predominant emphysema and a low postrehabilitation exercise capacity are most likely to benefit from LVRS.

Patients with an FEV_1 <20% of predicted and either diffusely distributed emphysema on CT scan or diffusing capacity of lung for carbon monoxide (D_{LCO}) <20% of predicted have increased mortality after the procedure, and thus are not candidates for LVRS.

Methods of achieving lung volume reduction by using bronchoscopic techniques have recently been approved by the U.S. Food and Drug Administration; they appear to be beneficial in selected emphysema patients.

Lung Transplantation (See also Chap. 298) COPD is currently the second leading indication for lung transplantation. Current recommendations are that candidates for lung transplantation should have very severe airflow obstruction, severe disability despite maximal medical therapy, and be free of significant comorbid conditions such as liver, renal, or cardiac disease.

EXACERBATIONS OF COPD

Exacerbations are a prominent feature of the natural history of COPD. Exacerbations are episodic acute worsening of respiratory symptoms, including increased dyspnea, cough, wheezing, and/or change in the amount and character of sputum. They may or may not be accompanied by other signs of illness, including fever, myalgias, and sore throat. The strongest single predictor of exacerbations is a history of a previous exacerbation. The frequency of exacerbations increases as airflow obstruction worsens; patients with severe (FEV_1 <50% predicted) or very severe airflow obstruction (FEV_1 <30% predicted) on average have 1–3 episodes per year. However, some individuals with very severe airflow obstruction do not have frequent exacerbations. Other factors, such as an elevated ratio of the diameter of the pulmonary artery to aorta on chest CT and gastroesophageal reflux, are also associated with increased risk of COPD exacerbations. Economic analyses have shown that >70% of COPD-related health care expenditures are due to emergency department visits and hospital care for COPD exacerbations; this translates to over $10 billion annually in the United States.

Precipitating Causes and Strategies to Reduce Frequency of Exacerbations A variety of stimuli may result in the final common pathway of airway inflammation and increased respiratory symptoms that are characteristic of COPD exacerbations. Studies suggest that acquiring a new strain of bacteria is associated with increased near-term risk of exacerbation and that bacterial infection/superinfection is involved in >50% of exacerbations. Viral respiratory infections are present in approximately one-third of COPD exacerbations. In a significant minority of instances (20–35%), no specific precipitant can be identified.

Patient Assessment An attempt should be made to establish the severity of the exacerbation as well as the severity of preexisting COPD. The more severe either of these two components, the more likely that the patient will require hospital admission. The history should include quantification of the degree and change in dyspnea by asking about breathlessness during activities of daily living and typical activities for the patient. The patient should be asked about fever; change in character of sputum; and associated symptoms such as wheezing, nausea, vomiting, diarrhea, myalgias, and chills. Inquiring about the frequency and severity of prior exacerbations can provide important information; the single greatest risk factor for hospitalization with an exacerbation is a history of previous hospitalization.

The physical examination should incorporate an assessment of the degree of distress of the patient. Specific attention should be focused on tachycardia, tachypnea, use of accessory muscles, signs of perioral or peripheral cyanosis, the ability to speak in complete

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sentences, and the patient's mental status. The chest examination should establish the presence or absence of focal findings, degree of air movement, presence or absence of wheezing, asymmetry in the chest examination (suggesting large airway obstruction or pneumothorax mimicking an exacerbation), and the presence or absence of paradoxical motion of the abdominal wall.

Patients with severe underlying COPD, who are in moderate or severe distress, or those with focal findings should have a chest x-ray or chest CT scan. Approximately 25% of x-rays in this clinical situation will be abnormal, with the most frequent findings being pneumonia and congestive heart failure. Patients with advanced COPD, a history of hypercarbia, or mental status changes (confusion, sleepiness) or those in significant distress should have an arterial blood gas measurement. The presence of hypercarbia, defined as a P_{CO_2} >45 mmHg, has important implications for treatment (discussed below). In contrast to its utility in the management of exacerbations of asthma, measurement of pulmonary function has not been demonstrated to be helpful in the diagnosis or management of exacerbations of COPD. Pulmonary embolus (PE) should also be considered, as the incidence of PE is increased in COPD exacerbations.

The need for inpatient treatment of exacerbations is suggested by the presence of respiratory acidosis and hypercarbia, new or worsening hypoxemia, severe underlying disease, and those whose living situation is not conducive to careful observation and the delivery of prescribed treatment.

TREATMENT OF ACUTE EXACERBATIONS

Bronchodilators Typically, patients are treated with inhaled beta agonists and muscarinic antagonists. These may be administered separately or together, and the frequency of administration depends on the severity of the exacerbation. Patients are often treated initially with nebulized therapy, as such treatment is often easier to administer in those in respiratory distress. It has been shown, however, that conversion to metered-dose inhalers is effective when accompanied by education and training of patients and staff. This approach has significant economic benefits and also allows an easier transition to outpatient care. The addition of methylxanthines (theophylline) to this regimen can be considered, although convincing proof of its efficacy is lacking. If methylxanthines are added, serum levels should be monitored in an attempt to minimize toxicity.

Antibiotics Patients with COPD are frequently colonized with potential respiratory pathogens, and it is often difficult to identify conclusively a specific species of bacteria responsible for a particular clinical event. Bacteria frequently implicated in COPD exacerbations include *Streptococcus pneumoniae, Haemophilus influenzae, Moraxella catarrhalis,* and *Chlamydia pneumoniae;* viral pathogens are also common etiologies of exacerbations. The choice of antibiotic should be based on local patterns of antibiotic susceptibility of the above bacterial pathogens as well as the patient's clinical condition. Patients with moderate or severe exacerbations are usually treated with antibiotics, even in the absence of data implicating a specific pathogen.

In patients admitted to the hospital, the use of systemic glucocorticoids reduces the length of stay, hastens recovery, and reduces the chance of subsequent exacerbation or relapse. One study demonstrated that 2 weeks of glucocorticoid therapy produced benefit indistinguishable from 8 weeks of therapy. Current recommendations suggest 30–40 mg of oral prednisolone or its equivalent typically for a period of 5–10 days in outpatients. Hyperglycemia, particularly in patients with preexisting diagnosis of diabetes, is the most frequently reported acute complication of glucocorticoid treatment.

Oxygen Supplemental O_2 should be supplied to maintain oxygen saturation ≥90%. Studies have demonstrated that in patients with both acute and chronic hypercarbia, the administration of supplemental O_2 does not reduce minute ventilation. It does, in some patients, result in modest increases in arterial P_{CO_2}, chiefly by altering ventilation-perfusion relationships within the lung. This should not deter practitioners from providing the oxygen needed to correct hypoxemia.

Mechanical Ventilatory Support The initiation of noninvasive positive-pressure ventilation (NIPPV) in patients with respiratory failure, defined as P_{aCO_2} >45 mmHg, results in a significant reduction in mortality rate, need for intubation, complications of therapy, and hospital length of stay. Contraindications to NIPPV include cardiovascular instability, impaired mental status, inability to cooperate, copious secretions or the inability to clear secretions, craniofacial abnormalities or trauma precluding effective fitting of mask, extreme obesity, or significant burns.

Invasive (conventional) mechanical ventilation via an endotracheal tube is indicated for patients with severe respiratory distress despite initial therapy, life-threatening hypoxemia, severe hypercarbia and/or acidosis, markedly impaired mental status, respiratory arrest, hemodynamic instability, or other complications. The goal of mechanical ventilation is to correct the aforementioned conditions. Factors to consider during mechanical ventilatory support include the need to provide sufficient expiratory time in patients with severe airflow obstruction and the presence of auto-PEEP (positive end-expiratory pressure), which can result in patients having to generate significant respiratory effort to trigger a breath during a demand mode of ventilation. The mortality rate of patients requiring mechanical ventilatory support is 17–30% for that particular hospitalization. For patients aged >65 admitted to the intensive care unit for treatment, the mortality rate doubles over the next year to 60%, regardless of whether mechanical ventilation was required.

Following a hospitalization for COPD, about 20% of patients are rehospitalized in the subsequent 30 days and 45% are hospitalized in the next year. Mortality following hospital discharge is about 20% in the following year.

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